

Utkarsh Mishra

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Education

SRM University

Bachelor of Technology

Electronics and Instrumentation Engineering

CGPA: 9.12

Chennai, TN

May. 2028

Experience

IIT Madras and KapiDhwaj Dynamics

Project Intern

Chennai, TN(on site)

Sep 2025 – Present

- Working on Computer vision, Deep learning, Robotic Operating system.
- Edge AI and multi-device systems for a robust architecture.
- Modern SOC's Nvidia Jetson Orin agx, Raspberry Pi5, STM32 and ESP32.
- Training Yolo models on a dataset for resources like roboflow.
- MATLAB for kinematic analysis.
- Electrical Subsystems, NEMA34 and YKD2305M stepper driver

Climec Labs

Creative Lead

intern

Chennai, TN(on site)

Aug-2024 – Sep-2025

- Worked on 3D Rendering and product placement in environments for advertisements.
- Using Blender and Premier Pro, for overall production of content and advertisements.
- Worked closely with founders discussing new ideas.

OnlyScrews

Social Media Manager

Chennai, TN(on site)

Aug-2024 – Sep-2025

- Termed as social media manager but working closely with the founders in operations, HR and creatives.
- Worked on creating advertisement and content for leads and more sales.
- Meta ads and google ads analysis as feedback for the content pipeline.

Projects

Air Quality Monitoring

Multi-Device IOT

April 2025 – May 2025

- Created an IoT system using multiple ESP32 microcontrollers to monitor indoor air quality
- Implemented multiple sensors for real-time acquisition of environmental information
- Created a Three.js 3D interface for visual representation and room monitoring
- Created custom enclosures using Fusion 360 for effective positioning of sensors

Material Speaks in spikes
Neuromorphic Computing

Sep 2025 – Oct 2025

- Developed hardware-based LIF (Leaky Integrate and Fire) neuron utilizing analog electronics
- Acoustic signatures were collected using microphone and analyzed based on spike generation
- Performed k-NN classification using ESP32 for instant material classification
- Showcased low-energy biological inspired sensor implementation for edge computing

Project VIPER

4+1 DOF Articulated robotic Arm (*Ongoing*)

Feb 2026 – Present

- Custom design of 4+1 Degrees of Freedom robotic manipulator through Fusion 360 and 3D printing
- Inverse Kinematics algorithm developed using ikpy with ROS2 for motion planning.
- Teach-and-repeat system using manually guided potentiometers
- Autonomous pick and place algorithm using Computer vision (ArUco markers)
- Architecture involves Raspberry Pi 5 + ESP32

Campus Delivery Bot (*Team Project, Ongoing*)

Autonomous Vehicle for inter campus logistics

Nov 2025 – Present

- Designing an Autonomous Vehicle Based on ROS2 for Logistics within Campus Boundaries
- Using SLAM and Navigation Capabilities through NAV2 and Google Cartographer
- Utilizing Multisensor Perception by Using Lidar, Intel RealSense, and IMU Sensors
- Based on NVIDIA Jetson Nano with STM32 for Efficient Edge Computing

Microwave Antenna

2.4GHZ microwave patch antenna

March 2026 – April 2026

- Designed and simulated a microstrip patch antenna with a frequency of 2.4 GHz for wireless communication systems utilizing Ansys HFSS
- Adjusted antenna parameters and dimensions for precise resonance at the targeted frequency
- Used the Rogers substrate to enhance the antenna performance
- Ensured the correctness of the design by means of electromagnetic simulation for frequency and impedance verification
- Calculated antenna dimensions using standard design equations with calculators available on the internet.

Skills & Interests

Technical:

- Robotics & Simulation: ROS2, Gazebo, NVIDIA Isaac Sim
- Programming: Python, C++
- Embedded & Electronics: ESP32, STM32, LTspice
- CAD & Design: Fusion 360, KiCAD
- Control & Analysis: MATLAB, Scilab
- Simulation & RF Tools: Ansys HFSS

Language: English (Professional Proficiency), Hindi (Native)

Interests: Music Production (FL Studio), Beatboxing, Filmmaking & Video Editing (Adobe Premiere Pro), 3D Art (Blender)
Cricket